Inflation Dynamics - Computational Notebook

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Abstract

This computational notebook documents the data construction, model specifications, estimation workflows, and best-model reporting used to study U.S. quarterly inflation dynamics.

This project studies U.S. quarterly inflation dynamics by combining survey-based inflation expectations with several conditional-volatility models for inflation innovations.
The notebook is organized around five linked pieces.

1. **Data summary.** The data section constructs the effective inflation sample **1969Q2–2022Q4**, with **215 observations** and give statistical summary.
2. **Mean processes.** The mean-process menu includes Constant, ARX(1,1), ARX(2,1), and ARX(2,2) specifications. These define $\hat{\pi}_t$ and therefore the residual sequence passed to volatility models.
3. **GARCH-family volatility models.** The GARCH section estimates GARCH, GJR-GARCH, and EGARCH specifications with normal, standardized Student's t, and Gaussian-mixture innovations.
4. **Regime-switching GARCH.** The regime-switching section studies state-dependent mean dynamics with normal and standardized Student's t innovations.
5. **BEGE-GARCH models.** The BEGE section estimates five volatility specifications: Constant, Symmetric, InflationDeflation, BadGood and Full BEGE models.

- π_t: realized inflation.
- $\mathcal{F}_{t-1}$: information set available before inflation at time t is realized.
- SPF_{t-1}: professional forecast of inflation based on $\mathcal{F}_{t-1}$.
- $\hat{\pi}_t$: expected inflation, $E[\pi_t \texttt{ mid } \mathcal{F}_{t-1}]$.
- u_t: inflation innovation, defined as $u_t := \pi_t - \hat{\pi}_t$.
- u_t^+: positive residual component, $u_t^+ := u_t \cdot \mathbf{1}(u_t > 0)$.
- u_t^-: negative residual component, $u_t^- := u_t \cdot \mathbf{1}(u_t \leq 0)$.

The rest of the notebook uses this notation consistently across data summaries, likelihood definitions, estimation scripts, and reported best-model equations.

Computation Notebook

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1. Inflation Summary Statistics

I start with simple statistical summary by plotting inflation and professional forecast over time.

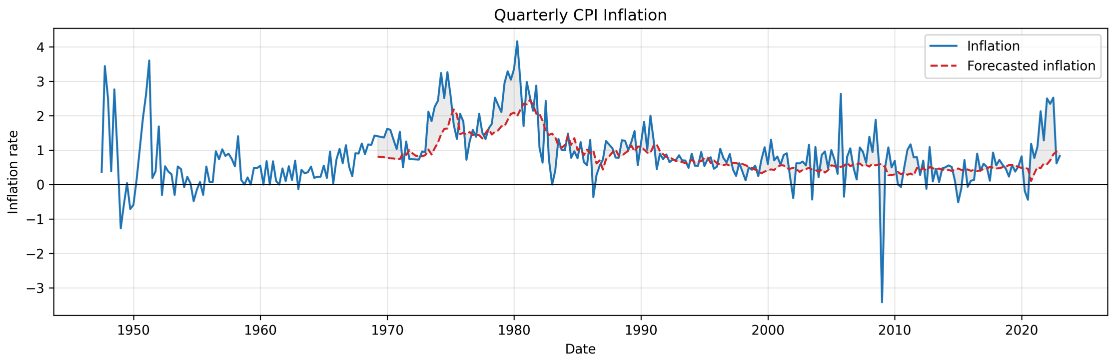


Figure 1: Quarterly inflation π_t and profesional forcast SPF_{t-1}.

Notice that inflation was recorded from 1947 but profesional forecast started from 1969.

	Inflation	Forecasted inflation
Date Start	1947Q2	1969Q1
Date End	2022Q4	2022Q4
#Observations	303	216

Table 1: Quaterly Inflation Statistics

1.a. Effective Sample

To make log-likelihood values comparable across specifications, I use an effective sample for all mean models and all volatility models.

In ARX models, lagged regressors (for example, SPF_{t-1}) are not observed at the very beginning of the raw sample. Therefore, I start the estimation sample at the first date where both current forecast inflation (SPF_t), and one-lag forecast inflation (SPF_{t-1}) are available. For the quarterly data, forecast inflation (SPF_t) is available from **1969Q1** to **2022Q4** (216 observations), requiring one lag (SPF_{t-1}) shifts the usable start to **1969Q2**. So the trimmed estimation window is **1969Q2–2022Q4**, with **215 observations**. This same trimmed sample is then used for all mean specifications, so the residual series has the same length across models.

For GARCH-type recursion, the first few conditional variances are unobserved (for example, σ_0^2, σ_{-1}^2, u_0^2, u_{-1}^2). I initialize these pre-sample variances using the model-implied unconditional variance, for example in GARCH model:

$$\bar{\sigma}^2 = \frac{\omega}{1 - \sum_i \alpha_i - \sum_j \beta_j}, \tag{1}$$

and set

$$u_0^2 = u_{-1}^2 = \cdots = \sigma_0^2 = \sigma_{-1}^2 = \cdots = \bar{\sigma}^2. \tag{2}$$

1.b. Effective Sample Summary Statistics

I report the summary statistics for trimmed effect sample. Skewness and Kurtosis are adjusted by sample size.

	Inflation (π)	SPF	$\pi - SPF$
Date Start	1969Q2	1969Q2	1969Q2
Date End	2022Q4	2022Q4	2022Q4
#Observations	215	215	215
Mean	0.9918	0.8320	0.1598
Median	0.7937	0.6296	0.1101
Std	0.8580	0.4969	0.6642
Min	−3.4170	0.1048	−4.0117
Max	4.1612	2.4566	2.1729
P5	−0.0681	0.3482	−0.7745
P25	0.5328	0.4748	−0.1669
P75	1.2913	1.0150	0.4617
P95	2.5883	2.0030	1.2581
Skewness	0.3550	1.3633	−0.7131
Kurtosis	3.8995	1.2053	7.3006
AC(1)	0.5981	0.9717	0.2764
AC(2)	0.5353	0.9473	0.1322
AC(4)	0.4708	0.8929	0.0881
AC(12)	0.2918	0.6949	0.0011

Table 2: Inflation statistics

2. Mean Process Specification

Four types of mean processes are of interest:

- **Constant**: $\pi_{t+1} = SPF_t + \mu_{t+1}$
- **ARX(1,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,1)**: $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \mu_{t+1}$
- **ARX(2,2)** : $\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 SPF_t + \phi_2 SPF_{t-1} + \mu_{t+1}$

where μ_{t+1} is the residual term.

2.a. OLS Results

Estimation results of four mean models with HAC standard errors in parentheses below each estimate. (The log-likelihood is computed under the Gaussian assumption.)

Constant:

$$\hat{\pi}_{t+1} = SPF_t \tag{3}$$

No estimated coefficients. Residuals $\mu_{t+1} = \pi_{t+1} - SPF_t$.

- Log-likelihood: −**222.664**
- AIC: **447.329**, BIC: **450.700**
- Residual skewness: −**0.713**, excess kurtosis: **7.301**

ARX(1,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0824} \; \underset{(0.112)}{+0.3005\,\pi_t} \; \underset{(0.167)}{+0.7337\,SPF_t} \tag{4}$$

- Log-likelihood: −**207.381**
- AIC: **420.762**, BIC: **430.874**
- Residual skewness: −**1.047**, excess kurtosis: **8.210**

ARX(2,1):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0897} \; \underset{(0.098)}{+0.2892\,\pi_t} \; \underset{(0.126)}{+0.0834\,\pi_{t-1}} \; \underset{(0.251)}{+0.6385\,SPF_t} \tag{5}$$

- Log-likelihood: −**206.810**
- AIC: **421.619**, BIC: **435.102**
- Residual skewness: −**1.191**, excess kurtosis: **9.050**

ARX(2,2):

$$\hat{\pi}_{t+1} = \underset{(0.086)}{0.0856} \; \underset{(0.106)}{+0.2992\,\pi_t} \; \underset{(0.125)}{+0.0914\,\pi_{t-1}} \; \underset{(0.433)}{+0.4084\,SPF_t} \; \underset{(0.297)}{+0.2136\,SPF_{t-1}} \tag{6}$$

- Log-likelihood: −**206.660**
- AIC: **423.319**, BIC: **440.172**
- Residual skewness: −**1.212**, excess kurtosis: **9.129**

For the following volatility estimation process, I impose the bounds on mean process parameters for stationarity. **Table 1: Parameter Bounds for Mean Process Specifications**

Model	c	ρ_1	ρ_2	ϕ_1	ϕ_2
Constant	–	–	–	–	–
ARX(1,1)	$(\min \pi_t,\ \max \pi_t)$	$(-1,\ 1)$	–	$(-10,\ 10)$	–
ARX(2,1)	$(\min \pi_t,\ \max \pi_t)$	$(-2,\ 2)$	$(-1,\ 1)$	$(-10,\ 10)$	–
ARX(2,2)	$(\min \pi_t,\ \max \pi_t)$	$(-2,\ 2)$	$(-1,\ 1)$	$(-10,\ 10)$	$(-10,\ 10)$

3. GARCH Family

This section presents three GARCH-type volatility models — GARCH, GJR-GARCH, and EGARCH — each combined with one of three conditional distributions for residuals: normal, Student's t, and a finite mixture of normals.

Let $\{u_t\}_{t=1}^{T}$ denote the residual process, and let $\mathcal{F}_{t-1}$ denote the information set available at time $t-1$. We assume

$$u_t = \sigma_t\, z_t, \qquad z_t \mid \mathcal{F}_{t-1} \overset{\text{i. i. d.}}{\sim} D(0,1), \tag{7}$$

where $\sigma_t^2 = \operatorname{Var}(u_t \mid \mathcal{F}_{t-1})$ is the conditional variance and $D(0,1)$ is a standardized distribution (normal, Student's t, or mixture of normals) with zero mean and unit variance. Throughout, $z_t = u_t / \sigma_t$ denotes the standardized residual and θ denotes the full parameter vector (mean process, volatility process parameters together with any distributional parameters).

3.a. *Volatility Recursion Specifications*

GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\, u_{t-i}^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{8}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\beta_k \geq 0$, and $\sum_i \alpha_i + \sum_k \beta_k < 1$ for stationarity.

GJR-GARCH(p, q):

$$\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i \left(u_{t-i}^{+}\right)^2 + \sum_{j=1}^{p} \gamma_j \left(u_{t-j}^{-}\right)^2 + \sum_{k=1}^{q} \beta_k\, \sigma_{t-k}^2, \tag{9}$$

with $\omega > 0$, $\alpha_i \geq 0$, $\gamma_j \geq 0$, and $\beta_k \geq 0$. The asymmetry coefficient γ_j captures the leverage effect: when $\alpha_j > \gamma_j$, positive shocks raise future volatility more than negative shocks of equal magnitude.

Under the assumption that z_t is symmetric

$$\mathbb{E}\left[\left(u_{t-i}^{+}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \mathbb{E}\left[\left(u_{t-i}^{-}\right)^2 \mid \mathcal{F}_{t-i-1}\right] = \frac{1}{2}\sigma_{t-i}^2 \tag{10}$$

, the stationarity condition is

$$\frac{1}{2}\sum_{i=1}^{p} \alpha_i + \frac{1}{2}\sum_{j=1}^{p} \gamma_j + \sum_{k=1}^{q} \beta_k < 1. \tag{11}$$

EGARCH(p, q):

$$\ln \sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i(\mid z_{t-i} \mid - \mathbb{E} \mid z_{t-i} \mid) + \sum_{j=1}^{p} \gamma_j\, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln \sigma_{t-k}^2, \tag{12}$$

where $z_t = u_t / \sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks), while γ_j captures the sign / leverage effect. Modeling $\ln \sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln \sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{13}$$

The constant $\mathbb{E} \mid z_t \mid$ depends on the assumed distribution of the standardized innovation:

- **Standard normal**, $z_t \sim \mathcal{N}(0, 1)$:

$$\mathbb{E} \mid z_t \mid = \sqrt{\frac{2}{\pi}}. \tag{14}$$

- **Standardized Student's t** with $\nu > 2$ degrees of freedom (rescaled to unit variance):

$$\mathbb{E} \mid z_t \mid = \frac{2\sqrt{\nu - 2}\,\Gamma\left(\frac{\nu+1}{2}\right)}{(\nu - 1)\sqrt{\pi}\,\Gamma\left(\frac{\nu}{2}\right)}. \tag{15}$$

As $\nu \to \infty$ this collapses to the normal value $\sqrt{2/\pi}$; for finite ν it is strictly larger, reflecting the heavier tails.

- **Mixture of two normals** with parameters $\left(p_1, \mu_1, \sigma_1^2\right)$ and induced $\left(p_2, \mu_2, \sigma_2^2\right)$ (as defined in the likelihood section): using $\mathbb{E} \mid X \mid = \mu\,[2\Phi(\mu/s) - 1] + 2s\,\phi(\mu/s)$ for $X \sim \mathcal{N}\left(\mu, s^2\right)$ (a folded-normal mean), where Φ and ϕ are the standard normal CDF and PDF,

$$\mathbb{E} \mid z_t \mid = \sum_{m=1}^{2} p_m[\,\mu_m(2\Phi(\mu_m/\sigma_m) - 1) + 2\sigma_m\,\phi(\mu_m/\sigma_m)]. \tag{16}$$

EGARCH(p, q):

$$\ln\sigma_t^2 = \omega + \sum_{i=1}^{p} \alpha_i\left(\mid z_{t-i} \mid - \sqrt{\frac{2}{\pi}}\right) + \sum_{j=1}^{p} \gamma_j\, z_{t-j} + \sum_{k=1}^{q} \beta_k \ln\sigma_{t-k}^2, \tag{17}$$

where $z_t = u_t/\sigma_t$ is the standardized residual. The term α_i captures the magnitude effect (response to the size of shocks) while γ_j captures the sign / leverage effect. Modeling $\ln\sigma_t^2$ guarantees positivity of σ_t^2 without sign restrictions on $\left(\alpha_i, \gamma_j, \beta_k\right)$, and the recursion is covariance-stationary in $\ln\sigma_t^2$ whenever

$$\sum_{k=1}^{q} \beta_k < 1. \tag{18}$$

Remark on the centering constant. The constant $\sqrt{2/\pi}$ equals $\mathbb{E} \mid z_t \mid$ when $z_t \sim \mathcal{N}(0, 1)$, so under a Gaussian innovation the term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ has zero conditional mean and ω admits the clean interpretation

$$\mathbb{E}[\ln\sigma_t^2] = \frac{\omega}{1 - \sum_{k=1}^{q} \beta_k}. \tag{19}$$

Following the convention of the `arch` Python package, we retain $\sqrt{2/\pi}$ as a fixed constant in the recursion even when the innovation distribution is Student's t or a mixture of normals. Under those distributions, $\mathbb{E} \mid z_t \mid \neq \sqrt{2/\pi}$, so the centered term $\mid z_{t-i} \mid - \sqrt{2/\pi}$ no longer has zero mean and ω loses the unconditional-mean interpretation above. This choice is observationally innocuous: replacing $\sqrt{2/\pi}$ with the distribution-specific constant $c_D = \mathbb{E}_D \mid z_t \mid$ yields an equivalent model whose fitted $\hat{\alpha}_i, \hat{\gamma}_j, \hat{\beta}_k$ are unchanged, with only the intercept shifted by

$$\omega = \omega^* + \left(\sqrt{\frac{2}{\pi}} - c_D\right) \sum_{i=1}^{p} \alpha_i. \tag{20}$$

The likelihood, the fitted σ_t^2 path, and all forecasts are identical under the two parameterizations.

3.b. Log-Likelihood Specifications

For each candidate distribution we give (i) the standardized density of z_t, (ii) the conditional density of $u_t \mid \mathcal{F}_{t-1}$ obtained via the change of variables $u_t = \sigma_t z_t$ (which contributes a Jacobian factor $1/\sigma_t$), (iii) the per-observation log-likelihood $\ell_t(\theta)$, and (iv) the sample log-likelihood

$$\ln L(\theta) = \sum_{t=1}^{T} \ell_t(\theta). \tag{21}$$

3.b.i. *Normal distribution:*

The standardized innovation satisfies $z_t \mid \mathcal{F}_{t-1} \sim \mathcal{N}(0, 1)$ with density

$$\phi(z) = \frac{1}{\sqrt{2\pi}} \exp\left\{-\frac{z^2}{2}\right\}. \tag{22}$$

By the change of variables $u_t = \sigma_t z_t$, the conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} \phi\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sqrt{2\pi\,\sigma_t^2}} \exp\left\{-\frac{u_t^2}{2\,\sigma_t^2}\right\}. \tag{23}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\frac{1}{2}\left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right), \tag{24}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\frac{1}{2} \sum_{t=1}^{T} \left(\ln 2\pi + \ln \sigma_t^2 + \frac{u_t^2}{\sigma_t^2}\right). \tag{25}$$

3.b.ii. *Student's t distribution:*

Let $\nu > 2$ denote the degrees-of-freedom parameter. The standardized Student's t density (rescaled to unit variance) is

$$f_\nu(z) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)}} \left(1 + \frac{z^2}{\nu-2}\right)^{-\frac{\nu+1}{2}}, \tag{26}$$

where $\Gamma(\cdot)$ is the gamma function. The restriction $\nu > 2$ ensures finite variance. The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1}; \theta) = \frac{1}{\sigma_t} f_\nu\left(\frac{u_t}{\sigma_t}\right) = \frac{\Gamma\left(\frac{\nu+1}{2}\right)}{\Gamma\left(\frac{\nu}{2}\right)\sqrt{\pi(\nu-2)\,\sigma_t^2}} \left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right)^{-\frac{\nu+1}{2}}. \tag{27}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = \ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\sigma_t^2\ \right) - \frac{\nu+1}{2}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right), \tag{28}$$

and the sample log-likelihood is

$$\ln L(\theta) = T\left[\ln\Gamma\left(\frac{\nu+1}{2}\right) - \ln\Gamma\left(\frac{\nu}{2}\right) - \frac{1}{2}\ln\left(\ \pi(\nu-2)\ \right)\right] - \frac{1}{2}\sum_{t=1}^{T}\ln\sigma_t^2 - \frac{\nu+1}{2}\sum_{t=1}^{T}\ln\left(1 + \frac{u_t^2}{(\nu-2)\,\sigma_t^2}\right). \tag{29}$$

3.b.iii. *Mixture of two normals:*

The standardized innovation follows a two-component Gaussian mixture with parameters (p_1, μ_1, σ_1^2):

$$f_{\text{mix}}(z) = p_1\,\phi(z;\mu_1,\sigma_1^2) + p_2\,\phi(z;\mu_2,\sigma_2^2), \qquad \phi(z;\mu,s^2) = \frac{1}{\sqrt{2\pi s^2}}\exp\left\{-\frac{(z-\mu)^2}{2s^2}\right\}. \tag{30}$$

To enforce $\mathbb{E}[z_t] = 0$ and $\mathrm{Var}(z_t) = 1$, the remaining parameters are determined by

$$p_2 = 1 - p_1, \qquad \mu_2 = -\frac{p_1\mu_1}{p_2}, \qquad \sigma_2^2 = \frac{1 - p_1(\sigma_1^2 + \mu_1^2) - p_2\mu_2^2}{p_2}. \tag{31}$$

This leaves three free parameters: (p_1, μ_1, σ_1^2).

Hierarchical representation. The mixture admits the latent-variable form $S \sim \text{Bernoulli}(p_1)$, $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$, $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$, with $z = X_1$ if $S = 1$ and $z = X_2$ otherwise. As a consequence, moments and the CDF decompose as

$$\mathbb{E}[z^n] = p_1\,\mathbb{E}[X_1^n] + p_2\,\mathbb{E}[X_2^n], \qquad F_{\text{mix}}(a) = p_1\,\Phi\left(\frac{a-\mu_1}{\sigma_1}\right) + p_2\,\Phi\left(\frac{a-\mu_2}{\sigma_2}\right), \tag{32}$$

where Φ is the standard normal CDF.

The conditional density of u_t is

$$f(u_t \mid \mathcal{F}_{t-1};\theta) = \frac{1}{\sigma_t}f_{\text{mix}}\left(\frac{u_t}{\sigma_t}\right) = \frac{1}{\sigma_t}\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \qquad z_t = \frac{u_t}{\sigma_t}. \tag{33}$$

The per-observation log-likelihood is

$$\ell_t(\theta) = -\ln\sigma_t + \ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right], \tag{34}$$

and the sample log-likelihood is

$$\ln L(\theta) = -\sum_{t=1}^{T}\ln\sigma_t + \sum_{t=1}^{T}\ln\left[p_1\,\phi(z_t;\mu_1,\sigma_1^2) + p_2\,\phi(z_t;\mu_2,\sigma_2^2)\right]. \tag{35}$$

3.b.iv. *Gaussian Mixture distribution:*

The mixture of 2 normal distribution given parameters p_1,μ_1,σ_1^2 is

$$f(z_t) = p_1\frac{1}{\sqrt{2\pi\sigma_1^2}}exp\left\{-\frac{\left(z_t-\mu_1\right)^2}{2\sigma_1^2}\right\} + p_2\frac{1}{\sqrt{2\pi\sigma_2^2}}exp\left\{-\frac{\left(z_t-\mu_2\right)^2}{2\sigma_2^2}\right\} \tag{36}$$

where $p_2 = 1 - p_1$,$\mu_2 = \frac{-p_1\mu_1}{p_2}$, and $\sigma_2^2 = \frac{1-p_2\mu_2^2-p_1(\sigma_1^2+\mu_1^2)}{p_2}$

N th order moment can be calculated as:

$$E[z^n] = \int z^n \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + (1-p_1) \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (37)$$

$$E[z^n] = p_1 E[x_1^n \mid x_1 \sim N(\mu_1, \sigma_1^2)] + p_2 E[x_2^n \mid x_2 \sim N(\mu_2, \sigma_2^2)] \quad (38)$$

Similary, CDF can be calculated analytically by

$$\Phi(a) = \int_{-\inf}^{a} z \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z-\mu_2)^2}{2\sigma_2^2} \right\} \right] dz \quad (39)$$

$$\Phi(a) = p_1 \Phi_{x_1}(a) + p_2 \Phi_{x_2}(a) \quad (40)$$

We can think mixture of two normal distributions generated by three random variable Z,X_1,X_2. Z follows Bernoulli(P_1), $X_1 \sim N(\mu_1, \sigma_1^2)$ and $X_2 \sim N(\mu_2, \sigma_2^2)$. In order to generate a random sample point, we can first use Z to decide which normal distributions to use, and then randomly pick a point in that normal distribution.

Log likelihood function is

$$L(u_t) = \sum_{t=1}^{T} \ln \frac{1}{\sigma_t} \left[p_1 \frac{1}{\sqrt{2\pi\sigma_1^2}} exp\left\{ -\frac{(z_t-\mu_1)^2}{2\sigma_1^2} \right\} + p_2 \frac{1}{\sqrt{2\pi\sigma_2^2}} exp\left\{ -\frac{(z_t-\mu_2)^2}{2\sigma_2^2} \right\} \right] \quad (41)$$

, where $z_t = \frac{u_t}{\sigma_t}$

4. Model Selection Report

Table 2: Model selection by AIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH AIC	GJR Mean	GJR Vol	GJR AIC	EGARCH Mean	EGARCH Vol	EGARCH AIC
Normal	(1,1)	(2,1)	387.6789	(1,1)	(2,1)	372.8425	(1,1)	(2,1)	374.2995
t	(2,1)	(2,1)	362.6718	(1,1)	(2,1)	357.3153	(1,1)	(2,1)	356.7451
Mix of Normal	(2,1)	(1,2)	365.6060	(2,1)	(2,1)	359.0690	(1,1)	(2,1)	356.4744

Table 3: Model selection by BIC: GARCH vs GJR vs EGARCH

Distribution	GARCH Mean	GARCH Vol	GARCH BIC	GJR Mean	GJR Vol	GJR BIC	EGARCH Mean	EGARCH Vol	EGARCH BIC
Normal	FC	(2,1)	401.4156	FC	(2,1)	396.6581	FC	(2,1)	398.1066
t	FC	(2,1)	387.7770	FC	(1,1)	384.2113	FC	(1,1)	383.8115
Mix of Normal	(1,1)	(2,1)	401.6779	FC	(2,1)	391.9670	FC	(1,1)	385.2567

4.a. Best Models by Criterion and Volatility Family

For each criterion and each volatility family, the selected model is the best across mean-process choices, orders, and distributions using stable & successful fits.

4.a.i. *AIC:*

GARCH:

- Model ID: `M079`
- Distribution: `$t$`
- Mean process: `(2,1)` (ARX_2_1)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0754 + 0.2410\,\pi_t + 0.1539\,\pi_{t-1} + 0.5650\,SPF_t + \mu_{t+1} \tag{42}$$

Robust SE: `Const` (0.0804), `Inflation_lag_1` (0.1565), `Inflation_lag_2` (0.2555), `SPF` (0.4729).

$$\hat{\sigma}_t^2 = 0.1369 + 0.4194\,u_{t-1}^2 + 0.3468\,u_{t-2}^2 + 0.0588\,\sigma_{t-1}^2 \tag{43}$$

- Number of observations: **215**
- Log-likelihood: **−172.335915**
- AIC: **362.671831**, BIC: **393.007573**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.075394	0.080383	0.937930	0.348280
Inflation_lag_1	0.241000	0.156521	1.539733	0.123625

Parameter	Coef	Std Err	t-value	p-value
Inflation_lag_2	0.153880	0.255527	0.602208	0.547036
SPF	0.565016	0.472887	1.194822	0.232156
alpha[1]	0.419351	0.260560	1.609418	0.107525
alpha[2]	0.346759	1.053861	0.329037	0.742128
beta[1]	0.058760	2.090575	0.028107	0.977577
nu	4.472871	1.614077	2.771164	0.005586
omega	0.136928	0.470167	0.291233	0.770873

GJR-GARCH:

- Model ID: `M068`
- Distribution: `$t$`
- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `GJR-GARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0846 + 0.1984\,\pi_t + 0.7847\,SPF_t + \mu_{t+1} \tag{44}$$

Robust SE: `Const` (0.0816), `Inflation_lag_1` (0.0996), `SPF` (0.1317).

$$\hat{\sigma}_t^2 = 0.1168 + 0.4252\left(u_{t-1}^{+}\right)^2 + 0.2174\left(u_{t-1}^{-}\right)^2 + 0.6575\left(u_{t-2}^{+}\right)^2 + 0.0000\left(u_{t-2}^{-}\right)^2 + 0.1354\,\sigma_{t-1}^2 \tag{45}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−168.657674**
- AIC: **357.315348**, BIC: **391.021728**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.084561	0.081554	1.036868	0.299797
Inflation_lag_1	0.198379	0.099598	1.991802	0.046393
SPF	0.784679	0.131672	5.959354	0.000000
alpha[1]	0.425191	0.219441	1.937612	0.052671
alpha[2]	0.657467	0.374661	1.754833	0.079288
beta[1]	0.135397	0.172961	0.782816	0.433735
gamma[1]	−0.207829	0.251448	−0.826528	0.408504
gamma[2]	−0.657467	0.305776	−2.150159	0.031543
nu	5.349660	2.200946	2.430618	0.015073
omega	0.116818	0.042038	2.778885	0.005455

EGARCH:

- Model ID: `M117`
- Distribution: `Mix of Normal`

- Mean process: `(1,1)` (ARX_1_1)
- Volatility process: `EGARCH(2,1)`

$$\hat{\pi}_{t+1} = 0.0861 + 0.2148\,\pi_t + 0.7325\,SPF_t + \mu_{t+1} \tag{46}$$

Robust SE: `Const` (0.0317), `Inflation_lag_1` (0.0705), `SPF` (0.0017).

$$\ln \hat{\sigma}_t^2 = -0.3146 + 0.5764\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.0053\left(\mid z_{t-2} \mid - \sqrt{2/\pi}\right) + 0.1006\,z_{t-1} + 0.3061\,z_{t-2} + 0.6993\ln \sigma_{t-1}^2 \tag{47}$$

- Number of observations: **215**
- Log-likelihood: **−166.237213**
- AIC: **356.474426**, BIC: **396.922083**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
Const	0.086145	0.031694	2.718014	0.006568
Inflation_lag_1	0.214824	0.070517	3.046416	0.002316
SPF	0.732514	0.001739	421.149922	0.000000
alpha[1]	0.576377	0.147631	3.904164	0.000095
alpha[2]	0.005343	0.238386	0.022413	0.982118
beta[1]	0.699319	0.105518	6.627490	0.000000
gamma[1]	0.100565	0.099839	1.007278	0.313801
gamma[2]	0.306148	0.099347	3.081593	0.002059
mu_1	0.048865	0.057085	0.855992	0.392002
omega	−0.314557	0.158733	−1.981672	0.047516
p_1	0.934951	0.041199	22.693639	0.000000
sigma_1_sq	0.625460	0.155173	4.030724	0.000056

4.a.ii. *BIC:*

GARCH:

- Model ID: `M055`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GARCH(2,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{48}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.1221 + 0.4616\,u_{t-1}^2 + 0.4956\,u_{t-2}^2 + 0.0000\,\sigma_{t-1}^2 \tag{49}$$

- Number of observations: **215**
- Log-likelihood: **−180.461891**
- AIC: **370.923781**, BIC: **387.776971**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.461596	0.179330	2.574004	0.010053
alpha[2]	0.495589	0.229219	2.162077	0.030612
beta[1]	0.000000	0.151356	0.000000	1.000000
nu	4.807722	1.636402	2.937984	0.003304
omega	0.122062	0.045238	2.698212	0.006971

GJR-GARCH:

- Model ID: `M050`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `GJR-GARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{50}$$

No mean-process coefficients are estimated in this anchored specification.

$$\hat{\sigma}_t^2 = 0.0636 + 0.7929\,\left(u_{t-1}^+\right)^2 + 0.1465\,\left(u_{t-1}^-\right)^2 + 0.4366\,\sigma_{t-1}^2 \tag{51}$$

Reported in README notation. (`arch` estimates the equivalent indicator form.)

- Number of observations: **215**
- Log-likelihood: **−178.679065**
- AIC: **367.358130**, BIC: **384.211320**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.792917	0.256260	3.094187	0.001974
beta[1]	0.436615	0.116958	3.733109	0.000189
gamma[1]	−0.646461	0.231689	−2.790208	0.005267
nu	4.760505	1.533912	3.103507	0.001912
omega	0.063611	0.025467	2.497746	0.012499

EGARCH:

- Model ID: `M051`
- Distribution: `$t$`
- Mean process: `FC` (Constant_anchor)
- Volatility process: `EGARCH(1,1)`

$$\hat{\pi}_{t+1} = SPF_t + \mu_{t+1} \tag{52}$$

No mean-process coefficients are estimated in this anchored specification.

$$\ln \hat{\sigma}_t^2 = -0.2520 + 0.5595\left(\mid z_{t-1} \mid - \sqrt{2/\pi}\right) + 0.2663\, z_{t-1} + 0.7882 \ln \sigma_{t-1}^2 \tag{53}$$

- Number of observations: **215**
- Log-likelihood: **−178.479161**

- AIC: **366.958322**, BIC: **383.811512**
- Optimizer success: **True**

Parameter	Coef	Std Err	t-value	p-value
alpha[1]	0.559468	0.139250	4.017715	0.000059
beta[1]	0.788161	0.061181	12.882450	0.000000
gamma[1]	0.266302	0.077913	3.417950	0.000631
nu	5.151108	1.764823	2.918768	0.003514
omega	−0.251989	0.105679	−2.384477	0.017103

4.b. EGARCH and Initialization Notes

- Effective sample is fixed at 215 observations for all models (`hold_back=0` with explicit lag regressors in the mean equation).
- In `arch`, EGARCH uses the package's centered term `|z|-sqrt(2/pi)` in the recursion for all distributions.
- Under non-Gaussian errors (e.g., Student's t or mixture), this is an intercept reparameterization; fitted dynamics and likelihood are still valid.
- The recursion start (`backcast`) is updated iteratively during estimation in this script: fit -> implied initial variance from current parameters -> refit.
- Stability is monitored using a persistence proxy (<1): GARCH uses `sum(alpha)+sum(beta)`, GJR uses `sum(alpha)+0.5*sum(gamma)+sum(beta)`, EGARCH uses `sum(beta)`.

5. Regime-Switching Inflation Models

Let the latent state be

$$s_t \in \{1, ..., K\}, \qquad \Pr(s_t = j \mid s_{t-1} = i) = p_{ij}. \tag{54}$$

I estimate several specifications that differ in **which blocks of parameters are allowed to switch with s_t**. Taking the ARX(2,2) model as an example,

$$\pi_{t+1} = c + \rho_1 \pi_t + \rho_2 \pi_{t-1} + \phi_1 \,\mathrm{SPF}_t + \phi_2 \,\mathrm{SPF}_{t-1} + \mu_{t+1}, \tag{55}$$

the parameters are grouped into three blocks, and each block can independently be made regime-dependent:

Block	State-dependent parameters
AR component	$c_{s_t},\ \rho_{1,s_t},\ \rho_{2,s_t}$
SPF component	$\phi_{1,s_t},\ \phi_{2,s_t}$
Shock distribution	$\sigma_{s_t},\ \nu_{s_t}$

where ν_{s_t} denotes the degrees of freedom when assuming standardized Student-t distribution for residuals. A given specification is defined by selecting any subset of these three blocks to be switching; parameters in blocks not selected are held constant across regimes.

Mean	Dist	K	Sw.AR	Sw.SPF	Sw.Var	Sw.nu	LogLik	AIC	BIC
Constant	Normal	2	N	N	Y	N	−186.63	381.26	394.743
Constant	Normal	3	N	N	Y	N	−180.655	379.311	409.646
Constant	Student t	2	N	N	Y	Y	−185.7	383.4	403.624
Constant	Student t	3	N	N	Y	Y	−180.389	384.778	425.226
ARX(1,1)	Normal	2	N	N	Y	N	−177.189	368.378	391.972
ARX(1,1)	Normal	2	Y	N	Y	N	−175.118	368.235	398.571
ARX(1,1)	Normal	2	Y	Y	Y	N	−175.092	370.185	403.891
ARX(1,1)	Normal	3	N	N	Y	N	−173.724	371.448	411.896
ARX(1,1)	Normal	3	Y	N	Y	N	−162.374	356.747	410.677
ARX(1,1)	Student t	2	N	N	Y	Y	−175.494	368.988	399.324
ARX(1,1)	Student t	2	Y	N	Y	Y	−162.972	347.945	385.022
ARX(1,1)	Student t	2	Y	Y	Y	Y	−162.961	349.921	390.369
ARX(1,1)	Student t	3	N	N	Y	Y	−173.091	376.182	426.742
ARX(1,1)	Student t	3	Y	N	Y	Y	−155.151	348.301	412.343
ARX(2,1)	Normal	2	N	N	Y	N	−176.216	368.431	395.396
ARX(2,1)	Normal	2	Y	N	Y	N	−174.942	371.884	408.961
ARX(2,1)	Normal	2	Y	Y	Y	N	−174.377	372.754	413.201
ARX(2,1)	Normal	3	N	N	Y	N	−171.151	368.301	412.119
ARX(2,1)	Normal	3	Y	N	Y	N	−156.72	351.44	415.482
ARX(2,1)	Student t	2	N	N	Y	Y	−173.425	366.849	400.556
ARX(2,1)	Student t	2	Y	N	Y	Y	−160.571	347.142	390.96
ARX(2,1)	Student t	2	Y	Y	Y	Y	−159.534	347.068	394.257
ARX(2,1)	Student t	3	N	N	Y	Y	−171.069	374.139	428.069
ARX(2,1)	Student t	3	Y	N	Y	Y	−150.126	344.253	418.407
ARX(2,2)	Normal	2	N	N	Y	N	−175.961	369.922	400.258
ARX(2,2)	Normal	2	Y	N	Y	N	−174.534	373.067	413.515
ARX(2,2)	Normal	2	Y	Y	Y	N	−174.155	376.31	423.499
ARX(2,2)	Normal	3	N	N	Y	N	−171.029	370.057	417.246
ARX(2,2)	Normal	3	Y	N	Y	N	−156.663	353.325	420.738
ARX(2,2)	Student t	2	N	N	Y	Y	−173.233	368.466	405.543
ARX(2,2)	Student t	2	Y	N	Y	Y	−160.567	349.135	396.324
ARX(2,2)	Student t	2	Y	Y	Y	Y	−157.588	347.176	401.107
ARX(2,2)	Student t	3	N	N	Y	Y	−170.754	375.507	432.808
ARX(2,2)	Student t	3	Y	N	Y	Y	−149.466	344.932	422.457

6. Best Regime-Switching Model

Selected by AIC among converged models.

6.a. Model Summary

- Model label: `ARX(2,1)_student_t_r3_arY_spfN`
- Mean process: `ARX(2,1)`
- Mean equation target: `Inflation`
- Intercept included: `Y`
- Intercept switches by regime: `Y`
- Error distribution: `Student t`
- #Regime: `3`
- Switching AR: `Y`
- Switching SPF: `N`
- Switching distribution variance: `Y`
- Switching nu: `Y`
- Sample: `1969Q2 to 2022Q4`
- Nobs: `215`
- LogLik: `-150.126`
- AIC: `344.253`
- BIC: `418.407`

6.b. Mean Model Specification

```
Inflation = c_(s_t) + beta_{Inflation_lag_1}(s_t) * Inflation_lag_1 +
beta_{Inflation_lag_2}(s_t) * Inflation_lag_2 + beta_{SPF} * SPF + u_t
```

Regime-specific mean coefficients:

regime	Intercept	Inflation_lag_1	Inflation_lag_2	SPF
0	0.872769	−0.54292	−0.238057	0.574951
1	0.381796	0.0492982	−0.0989449	0.574951
2	−0.0670625	0.26872	0.448281	0.574951

6.c. Mean Process Parameters

parameter	estimate
const[0]	0.872769
const[1]	0.381796
const[2]	−0.0670625
Inflation_lag_1[0]	−0.54292
Inflation_lag_1[1]	0.0492982
Inflation_lag_1[2]	0.26872
Inflation_lag_2[0]	−0.238057
Inflation_lag_2[1]	−0.0989449
Inflation_lag_2[2]	0.448281

parameter	estimate
SPF	0.574951

6.d. Distribution Parameters

parameter	estimate
sigma2[0]	0.0118058
sigma2[1]	0.612973
sigma2[2]	0.208789
nu[0]	183.071
nu[1]	2.39744
nu[2]	104.326

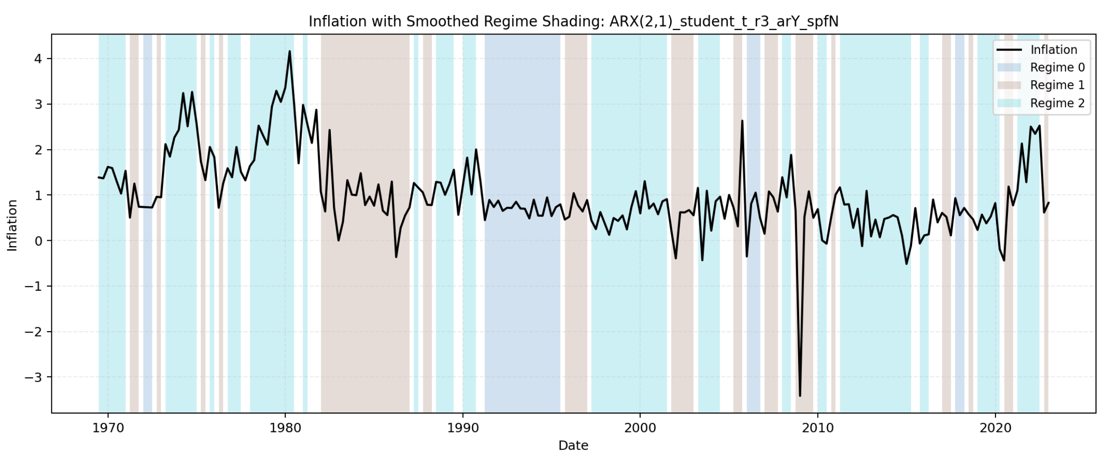
6.e. Transition Probabilities (Vector Form)

parameter	estimate
p[0->0]	0.832798
p[1->0]	0.0480447
p[2->0]	0.000150046
p[0->1]	0.165512
p[1->1]	0.646779
p[2->1]	0.24607

6.f. Transition Probability Matrix

From \ To	Regime 0	Regime 1	Regime 2	Row Sum
Regime 0	0.832798	0.165512	0.00168973	1
Regime 1	0.0480447	0.646779	0.305177	1
Regime 2	0.000150046	0.24607	0.75378	1

6.g. Regime Classification Plot



7. BEGE GARCH Family

To start the random search of initial mean parameters, I draw uniform samples from $(\mu - 2\sigma,\ \mu + 2\sigma)$ where μ and σ are mean and standard deviation from OLS regression. I also set the AR coefficient bound to avoid the AR process to explode. We have four types of mean processes.

7.a. Implied Moment Dynamics

The BEGE GARCH models have nice properties

- conditional variance: $\sigma_t^2 = \sigma_p^2 p_t + \sigma_n^2 n_t$,
- conditional skewness: $s_t^2 = 2\left(\sigma_p^3 p_t - \sigma_n^3 n_t\right)$,
- conditional excess kurtosis: $k_t^2 = 6\left(\sigma_p^4 p_t + \sigma_n^4 n_t\right)$.

Used for recursive constraints!!

7.b. Random Search

For the multi-start BEGE estimators, robust numerical standard errors are optional through `compute_se`. The default is `compute_se=False` for fast model search; the result collectors recompute standard errors for the reported top log-likelihood rows after selection.

7.c. Best-Model Reporting Screen

Raw BEGE search outputs are kept in `output/raw/draw_###.csv`. The collectors recompute the likelihood from each stored parameter vector using the stabilized BEGE density before writing `results/all_estimations.csv`, so stale likelihoods from earlier density code do not determine the best model. The cleaned estimations are also split by mean process under `results/by_mean/` as `constant.csv`, `ARX_1_1.csv`, `ARX_2_1.csv`, and `ARX_2_2.csv`; these by-mean files keep only rows with successful optimizer status and `selection_eligible=True`.

Reported best-model tables require finite corrected likelihood criteria, successful optimizer convergence, finite positive shape paths, positive conditional variance paths, mean-process stationarity, the documented parameter/stability constraints, and the EWMA implied-variance bounds applied to $\sigma_p^2 p_t + \sigma_n^2 n_t$.

The implied-variance bounds use the same screen during optimization and collection. For residuals from the effective sample, compute an EWMA path with $\lambda = 0.94$ and an initialization window of $\tau = \min(75, T)$. The lower and upper paths are

$$\max\left(EWMA_t/10^6,\ \operatorname{Var}(u_t)/10^8\right) \le \sigma_p^2 p_t + \sigma_n^2 n_t \le \max\left\{\min\left(EWMA_t \cdot 10^6,\ 10^7\left(1 + \max u_t^2\right)\right),\ 1 + \max u_t^2\right\}. \quad (56)$$

The diagnostic value $\max_t\{p_t, n_t\}$ is recorded but is not used as a selection exclusion rule. The companion `results/selection_diagnostics.csv` records stored versus corrected likelihood criteria, high-shape density usage, implied persistence, minimum scale, maximum shape path, implied-variance-bound status, mean stationarity status, and the exclusion reason for each saved estimation row. A log-likelihood value above `-150` is recorded as a manual review diagnostic but is not a selection exclusion rule.

Each `results/best_model.md` reports only the single likelihood-best admissible row for that BEGE specification. The markdown first shows the selected mean-process equation and BEGE volatility-process equation with the estimated parameters substituted directly into the equations; standard errors appear below the substituted estimates in parentheses. The same selected row with standard errors is written to `results/best_loglik_with_se.csv`.

8. BEGE Density

BEGE density function $f(u \mid p, n, \sigma_p, \sigma_n)$ is the function that calculates the density of the observation u given the parameters $\{p, n, \sigma_p, \sigma_n\}$. We have three code on table–Numerical Integration, Justin's code, and my improved code. I compare three code by computing the sum of log likelihood function on real residuals instead of sythetic data. This gives relative comparision between codes.

I compare three fixed-shape BEGE density implementations on the ARX(1,1) residuals from the canonical effective sample. As a benchmark, the Gaussian OLS log-likelihood is −207.381. The residuals are mean-zero (0.00) with a standard deviation of 0.636326, a negative skewness of −1.047441 and a large excess kurtosis of 8.210174.

8.a. Shape Parameter Range

The range below uses eligible ARX(1,1) rows from the BadGood BEGE results only. For each saved parameter vector I recomputed the recursive shape paths on the common ARX(1,1) residuals, then summarized all observations and all eligible rows. The density comparison itself keeps the selected shape values constant across time.

Model	Rows	p q05	p median	p q95	n q05	n median	n q95	sigma_p med	sigma_n med
BadGood	994	1.4285	2.8130	6.4330	0.0714	0.3320	2.8342	0.2355	0.5555

8.b. Fixed-Shape Parameter Sets

The log likelihood and timing comparison uses the following five BadGood-based parameter sets. In each row, `sigma_p` and `sigma_n` are held at their BadGood medians.

Set	p	n	sigma_p	sigma_n
p q05, n median	1.428538	0.331970	0.235491	0.555532
p median, n median	2.813004	0.331970	0.235491	0.555532
p q95, n median	6.433015	0.331970	0.235491	0.555532
p median, n q05	2.813004	0.071355	0.235491	0.555532
p median, n q95	2.813004	2.834194	0.235491	0.555532

8.c. Log Likelihood Comparison

`BEGE_density.py` is the current implementation. `BEGE_density_Justin.py` is Justin's analytic formula, and the numerical columns use `BEGE_density_Numerical_Integration.py::loglikedgam_constant` at the stated grid size.

Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	−214.162839	−214.162839	−223.900355	−213.330306	−214.116656	−214.148282	−214.146613	−214.160080

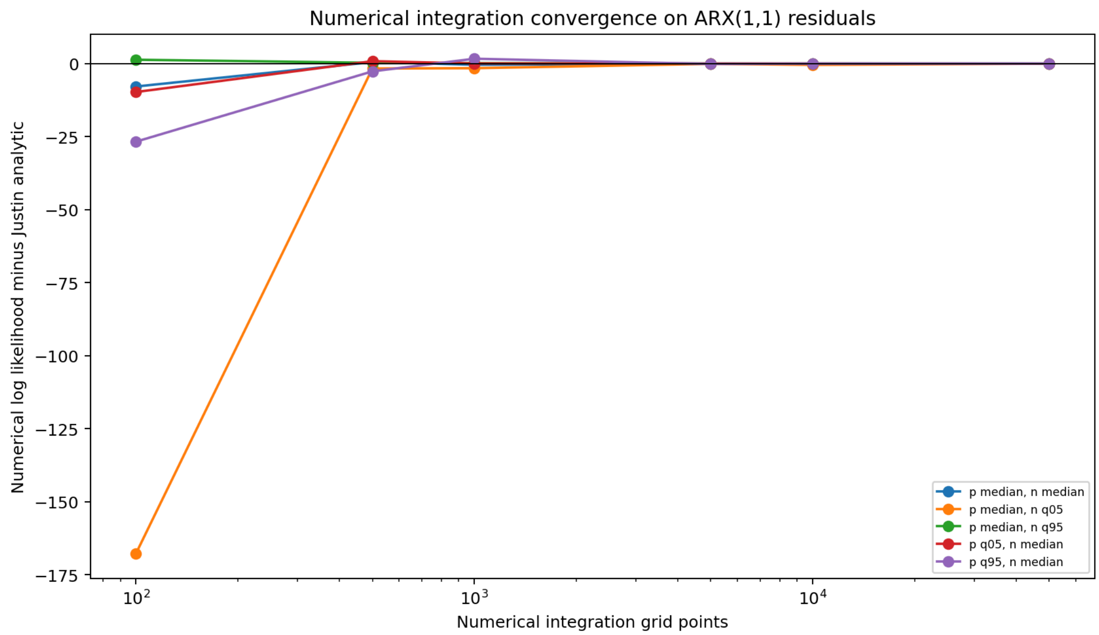
Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p median, n median	−188.957117	−188.957117	−196.801762	−188.438308	−189.359597	−189.033173	−188.947127	−188.954400
p q95, n median	−194.340481	−194.340481	−221.076698	−196.996227	−192.679189	−194.464527	−194.368904	−194.347495
p median, n q05	−212.877473	−212.877473	−380.654962	−214.556763	−214.448495	−212.976377	−213.323667	−212.904257
p median, n q95	−235.887717	−235.887717	−234.563988	−235.609633	−235.748009	−235.859932	−235.873999	−235.885265

- The old numerical integration method converges toward Justin's likelihood function as the number of grid points increases, but the convergence is slow and requires very fine grids.
- For small or moderate grid sizes, the numerical integration *overestimates* the log-likelihood.

8.d. Evaluation Speed Comparison

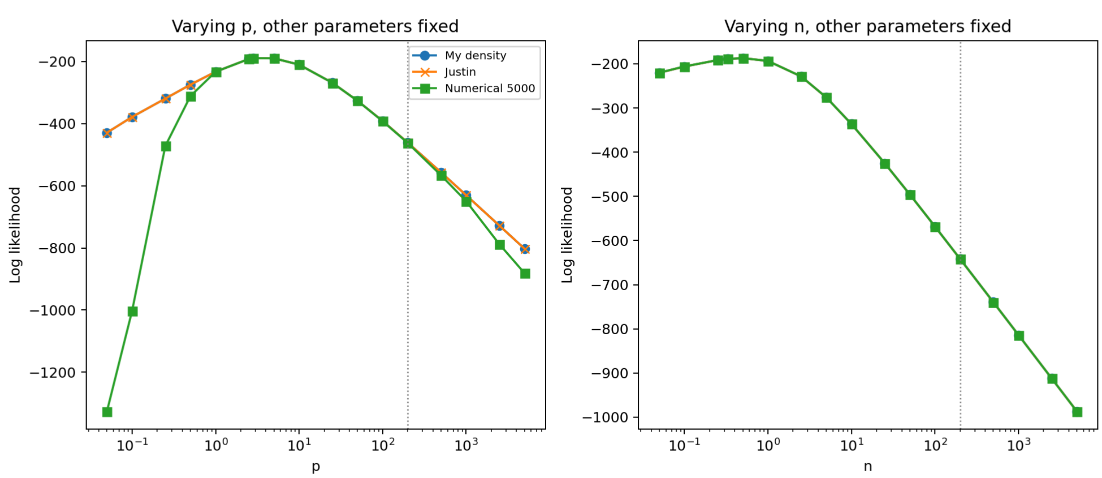
Parameter set	My function	Justin function	Numerical 100	Numerical 500	Numerical 1000	Numerical 5000	Numerical 10000	Numerical 50000
p q05, n median	0.0003	0.0005	0.0086	0.0427	0.0883	0.4326	0.9102	4.5478
p median, n median	0.0004	0.0006	0.0075	0.0358	0.0730	0.3674	0.7613	3.8117
p q95, n median	0.0003	0.0005	0.0064	0.0308	0.0663	0.3258	0.6520	3.2415
p median, n q05	0.0003	0.0005	0.0076	0.0358	0.0737	0.3567	0.7479	3.6527
p median, n q95	0.0004	0.0006	0.0059	0.0284	0.0611	0.2900	0.5937	2.9724

- My improved code provides an equavalent accurate and more computationally efficient evaluation.



8.e. Shape Tail Consistency

Holding the other parameters at the BadGood medians (`p=2.8130`, `n=0.3320`, `sigma_p=0.2355`, `sigma_n=0.5555`), I vary either `p` or `n` up to 5000 and compare the three density functions. The numerical integration line uses 5,000 grid points.



8.f. Findings

- BadGood quantiles provide the fixed-shape examples used in the likelihood and timing tables.
- The numerical integration function is useful as a convergence diagnostic, but it is much slower than the analytic functions and remains grid-sensitive.
- The shape-tail plot varies one shape parameter at a time while holding the other shape and both scale parameters at the BadGood medians.

9. Constant BEGE

BEGE with invariant shape parameters is

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1} \\ u_{t+1} &= \sigma_p \omega_p - \sigma_n \omega_n \end{aligned} \tag{57}$$

where

$$\omega_p \sim \tilde{\Gamma}(\bar{p}, 1), \quad \omega_n \sim \tilde{\Gamma}(\bar{n}, 1). \tag{58}$$

$\bar{p}$ and $\bar{n}$ are unconditional shape parameters. The random search and bounds are set to be:

- $10^{-5} < \sigma_p, \sigma_n < 2$,
- $0.1 < p, n < 10$.

The constant implied variance path

$$\sigma_p^2 \bar{p} + \sigma_n^2 \bar{n} \tag{59}$$

is screened against the same project EWMA lower and upper bounds used by the dynamic BEGE specifications during optimization and result collection.

10. Result Collection

`collect_constant_results.py` merges the raw seed files, writes the cleaned combined output to `results/all_estimations.csv`, and splits the same cleaned output by mean process under `results/by_mean/` as `constant.csv`, `ARX_1_1.csv`, `ARX_2_1.csv`, and `ARX_2_2.csv`. It ranks admissible estimates by log likelihood, computes standard errors for the likelihood-best admissible fit in `results/best_loglik_with_se.csv`, and writes `results/best_model.md` with only that selected fit as substituted mean and fixed-shape BEGE equations. Standard errors are shown below the parameter values in those reported equations.

11. Constant BEGE Best Model Summary

Generated: `2026-06-03T14:41:47` Total estimations: `8000` Successful estimations: `7997` Eligible estimations for best-model selection: `7997`

Selection screen: successful optimizer status, finite positive BEGE parameters, documented parameter bounds, EWMA implied-variance bounds, positive conditional variance, and mean-process stationarity. Log likelihoods above `-150` are flagged for review but are not excluded by this threshold. This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage for this selected model.

11.a. Selected Best Model

Best admissible estimate ranked by log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC	Implied Var	Above −150 Diagnostic
ARX(2,2)	18	1	−181.6884	381.3768	411.7126	0.3952	no

Selection checks:

- Optimizer convergence: `yes`
- Parameter bounds: `yes`
- BEGE stability and variance restrictions: `not applicable for fixed-shape Constant BEGE`
- Shape upper-cap diagnostic: `not applicable for fixed-shape Constant BEGE`
- Implied variance bounds: `yes`
- Mean-process stationarity: `yes`
- Selection diagnostics: `eligible`
- Standard errors: `computed`

Mean process:

$$\pi_{t+1} = \underset{(0.1082)}{0.0042} + \underset{(0.1196)}{0.3232}\,\pi_t + \underset{(0.0834)}{0.1633}\,\pi_{t-1} + \underset{(0.3064)}{0.4107}\,SPF_t + \underset{(0.2891)}{0.1942}\,SPF_{t-1} + u_{t+1} \tag{60}$$

BEGE volatility process:

$$\begin{aligned} u_t &= \underset{(0.0368)}{0.2712}\,\omega_{p,t} - \underset{(0.3484)}{0.8404}\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(\bar{p},1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(\bar{n},1), \\ \bar{p} &= \underset{(0.4671)}{2.6648}, \qquad \bar{n} = \underset{(0.1606)}{0.2821}, \\ \mathrm{Var}_t(u_t) &= \left(\underset{(0.0368)}{0.2712}\right)^2 \underset{(0.4671)}{2.6648} + \left(\underset{(0.3484)}{0.8404}\right)^2 \underset{(0.1606)}{0.2821}. \end{aligned} \tag{61}$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0042	0.1082
rho_1	0.3232	0.1196
rho_2	0.1633	0.0834

Parameter	Estimate	Std. Error
phi_1	0.4107	0.3064
phi_2	0.1942	0.2891
shape_p	2.6648	0.4671
shape_n	0.2821	0.1606
sigma_p	0.2712	0.0368
sigma_n	0.8404	0.3484

12. Symmetric BEGE

The model assumes different scaled parameters and different constants in GJR recursions for both components:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \, \omega_p - \sigma_n \, \omega_n, \end{aligned} \tag{62}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{63}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho, \phi^+, \phi^-)$:

$$\begin{aligned} p_t &= p_0 + \rho \, p_{t-1} + \frac{\phi^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho \, n_{t-1} + \frac{\phi^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2 \end{aligned} \tag{64}$$

The parameter bounds are chosen to be:

- $0 \le p_0, n_0 < 10$,
- $0 \le \rho \le 1$,
- $0 \le \phi^+, \phi^- \le 2$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

The current estimation code enforces the stability condition

$$\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1. \tag{65}$$

The optimizer and collector also enforce the EWMA implied-variance bounds on $\sigma_p^2 p_t + \sigma_n^2 n_t$ at every effective-sample observation.

13. Batch Estimation

`BEGE_symmetric1.py` estimates the Symmetric BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_symmetric_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/by_mean/constant.csv`, `results/by_mean/ARX_1_1.csv`, `results/by_mean/ARX_2_1.csv`, and `results/by_mean/ARX_2_2.csv`, which split the cleaned estimations by mean process.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_loglik_with_se.csv`, with standard errors for the likelihood-best admissible fit.
- `results/best_model.md`, with only the likelihood-best admissible fit, reported as substituted mean and BEGE volatility equations with standard errors shown below the parameter values.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationSymmetric.sh` defaults to 50 seed jobs, 40 draws per mean process, 25 optimizer starts per draw, and 800 SLSQP iterations per start. This gives 50,000 optimizer starts for each mean process and Symmetric BEGE specification. With all four mean processes, that is 200,000 optimizer starts total.

14. Symmetric BEGE Best Model Summary

Generated: `2026-06-03T14:51:48` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `7999`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, EWMA implied-variance bounds, mean-process stationarity, and documented parameter/stability/unconditional-variance constraints. Corrected log likelihoods above `-150` are flagged for review but are not excluded by this threshold. This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage for this selected model.

> **Note**
>
> Flagged 54 estimate(s) with corrected log likelihood above `-150` for manual review. These rows remain eligible if they pass the admissibility checks.

14.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC	Max Shape	Max Implied Var	Above −150 Diagnostic
ARX(2,1)	21	40	253.4762	−484.9524	−447.8754	271.1604	23.4717	yes

Selection checks:

- Optimizer convergence: `yes`
- Parameter bounds: `yes`
- BEGE stability and variance restrictions: `yes`
- Shape upper-cap diagnostic: `flagged`
- Implied variance bounds: `yes`
- Mean-process stationarity: `yes`
- Selection diagnostics: `eligible`
- Standard errors: `computed`

Mean process:

$$\pi_{t+1} = \underset{(30.1515)}{-0.0279} + \underset{(1.2296)}{0.4083}\,\pi_t + \underset{(20.4300)}{-0.0331}\,\pi_{t-1} + \underset{(26.7291)}{0.6653}\,SPF_t + u_{t+1} \tag{66}$$

BEGE volatility process:

$$\begin{gathered} u_t = \underset{(1.2348)}{0.0761}\,\omega_{p,t} - \underset{(4.3916)}{0.4957}\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{67}$$

$$
\begin{aligned}
p_t &= \underset{(66.6572)}{8.1612} + \underset{(0.0000)}{0.9400}\, p_{t-1} + \frac{\underset{(0.0698)}{0.0506}}{2\left(\underset{(1.2348)}{0.0761}\right)^2}\left(u_{t-1}^{+}\right)^2 + \frac{\underset{(1.1648)}{0.0093}}{2\left(\underset{(1.2348)}{0.0761}\right)^2}\left(u_{t-1}^{-}\right)^2, \\
n_t &= \underset{(6.9573)}{2.6820} + \underset{(0.0000)}{0.9400}\, n_{t-1} + \frac{\underset{(0.0698)}{0.0506}}{2\left(\underset{(4.3916)}{0.4957}\right)^2}\left(u_{t-1}^{+}\right)^2 + \frac{\underset{(1.1648)}{0.0093}}{2\left(\underset{(4.3916)}{0.4957}\right)^2}\left(u_{t-1}^{-}\right)^2
\end{aligned}
\tag{68}
$$

Parameter table:

Parameter	Estimate	Std. Error
c	−0.0279	30.1515
rho_1	0.4083	1.2296
rho_2	−0.0331	20.4300
phi_1	0.6653	26.7291
p0	8.1612	66.6572
n0	2.6820	6.9573
rho	0.9400	0.0000
phi_plus	0.0506	0.0698
phi_minus	0.0093	1.1648
sigma_p	0.0761	1.2348
sigma_n	0.4957	4.3916

15. Bad and Good Symmetric GARCH Model

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p, \phi_n\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p \, p_{t-1} + \frac{\phi_p}{2\sigma_p^2} \left(u_{t-1}\right)^2, \\ n_t &= n_0 + \rho_n \, n_{t-1} + \frac{\phi_n}{2\sigma_n^2} \left(u_{t-1}\right)^2 \end{aligned} \tag{69}$$

I have set the constraints below.

- $\rho + \phi < 1$ for both good and bad shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- The implied variance path $\sigma_p^2 p_t + \sigma_n^2 n_t$ must satisfy the project EWMA lower and upper bounds at every effective-sample observation.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization or reported best-model selection by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite; large shape states use the saddlepoint density backend. The collector writes the maximum shape path as a diagnostic in `results/selection_diagnostics.csv`.

16. Estimation Workflow

`BG_GJR1.py` estimates the BadGood BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_bg_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/by_mean/constant.csv`, `results/by_mean/ARX_1_1.csv`, `results/by_mean/ARX_2_1.csv`, and `results/by_mean/ARX_2_2.csv`, which split the cleaned estimations by mean process.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_loglik_with_se.csv`, with standard errors for the likelihood-best admissible fit.
- `results/best_model.md`, with only the likelihood-best admissible fit, reported as substituted mean and BEGE volatility equations with standard errors shown below the parameter values.

The by-mean CSV files keep only estimates with `optimizer_success=True` and `selection_eligible=True`; excluded raw-search rows remain in `results/all_estimations.csv` and `results/selection_diagnostics.csv`.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationBG.sh` defaults to 50 seed jobs, 40 draws per mean process, 25 optimizer starts per draw, and 800 SLSQP iterations per start. This gives 50,000 optimizer starts for each mean process and BadGood BEGE specification. With all four mean processes, that is 200,000 optimizer starts total. The `estimate` action prints a rough per-seed time estimate before submitting jobs.

17. BadGood BEGE Best Model Summary

Generated: `2026-06-03T14:38:52` Total estimations: `8000` Converged estimations: `7841` Eligible estimations for best-model selection: `7841`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, EWMA implied-variance bounds, mean-process stationarity, and documented parameter/stability/unconditional-variance constraints. Corrected log likelihoods above `-150` are flagged for review but are not excluded by this threshold. This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage for this selected model.

> **Note**
>
> Flagged 107 estimate(s) with corrected log likelihood above `-150` for manual review. These rows remain eligible if they pass the admissibility checks.

17.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC	Max Shape	Max Implied Var	Above −150 Diagnostic
ARX(1,1)	16	24	134.0336	−246.0672	−208.9902	3024.7460	11.5915	yes

Selection checks:

- Optimizer convergence: `yes`
- Parameter bounds: `yes`
- BEGE stability and variance restrictions: `yes`
- Shape upper-cap diagnostic: `flagged`
- Implied variance bounds: `yes`
- Mean-process stationarity: `yes`
- Selection diagnostics: `eligible`
- Standard errors: `computed`

Mean process:

$$\pi_{t+1} = \underset{(0.1663)}{0.3789} + \underset{(0.1689)}{0.4455}\,\pi_t + \underset{(0.3631)}{1.1357}\,SPF_t + u_{t+1} \tag{70}$$

BEGE volatility process:

$$\begin{aligned} u_t &= \underset{(3.3997)}{0.2718}\,\omega_{p,t} - \underset{(0.0002)}{0.0490}\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \tag{71}$$

$$
\begin{aligned}
p_t &= \underset{(3.2515)}{4.2462} + \underset{(0.0000)}{0.9213}\, p_{t-1} + \frac{\underset{(0.1315)}{0.0220}}{2\left(\underset{(3.3997)}{0.2718}\right)^2} u_{t-1}^2, \\
n_t &= \underset{(610.0596)}{9.6817} + \underset{(0.0000)}{0.2328}\, n_{t-1} + \frac{\underset{(2.9322)}{0.6270}}{2\left(\underset{(0.0002)}{0.0490}\right)^2} u_{t-1}^2
\end{aligned}
\tag{72}
$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.3789	0.1663
rho_1	0.4455	0.1689
phi_1	1.1357	0.3631
p0	4.2462	3.2515
n0	9.6817	610.0596
rho_p	0.9213	0.0000
rho_n	0.2328	0.0000
phi_p	0.0220	0.1315
phi_n	0.6270	2.9322
sigma_p	0.2718	3.3997
sigma_n	0.0490	0.0002

18. Inflation and Deflation Model

Here n_t, p_t is generated recursively with parameters $\left(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_n^-\right)$:

$$\begin{aligned} p_t &= p_0 + \rho_p\, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2}\left(u_{t-1}^+\right)^2, \\ n_t &= n_0 + \rho_n\, n_{t-1} + \frac{\phi_n^-}{2\sigma_n^2}\left(u_{t-1}^-\right)^2 \end{aligned} \tag{73}$$

For the current random-search estimation runs, the likelihood is evaluated under the parameter bounds used in code, finite-recursion checks, and the documented ID-GARCH stability restrictions:

- $\rho_p + \frac{\phi_p^+}{2} < 1$ and $\rho_n + \frac{\phi_n^-}{2} < 1$.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$.
- The implied variance path $\sigma_p^2 p_t + \sigma_n^2 n_t$ must satisfy the project EWMA lower and upper bounds at every effective-sample observation.

The hard shape cap $\max\{p_t, n_t\} < 200$ is **not imposed by default** in `ID_GARCH`. It can still be restored for sensitivity checks by passing `cap_pn=200`, but it is not part of the current best-model reporting screen.

For reported best-model selection, `collect_id_results.py` keeps the raw search rows, recomputes each likelihood with the stabilized BEGE density, and uses finite corrected criteria, optimizer convergence, and the documented parameter/stability/unconditional-variance constraints. The row-level selection outcome, stored likelihood, corrected likelihood, and maximum shape diagnostic are written to `results/selection_diagnostics.csv`.

19. Estimation Workflow

The Slurm workflow in `SimulationID.sh` submits 50 seed jobs by default. Each seed estimates all four mean processes on the fixed effective sample `1969Q2--2022Q4`:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

The job runner skips standard-error calculations and saves one CSV row after each model fit so partial jobs leave recoverable output. Each seed uses 40 draws per mean process and 25 optimizer starts per draw, giving 50,000 optimizer starts for each mean process and Inflation/Deflation BEGE specification. The collector merges the raw CSV files, writes the cleaned combined output to `results/all_estimations.csv`, and writes admissible cleaned outputs split by mean process under `results/by_mean/` as `constant.csv`, `ARX_1_1.csv`, `ARX_2_1.csv`, and `ARX_2_2.csv`. The by-mean CSV files keep only rows with `optimizer_success=True` and `selection_eligible=True`. It ranks admissible estimates by corrected log likelihood, computes standard errors for the likelihood-best admissible fit in `results/best_loglik_with_se.csv`, and writes `results/best_model.md` with only that selected fit as substituted mean and BEGE volatility equations. Standard errors are shown below the parameter values in those reported equations.

20. Inflation/Deflation BEGE-GJR Best Model Summary

Generated: `2026-06-03T14:44:25` Total estimations: `8000` Converged estimations: `7310` Eligible estimations for best-model selection: `7273`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, EWMA implied-variance bounds, mean-process stationarity, and documented parameter/stability/unconditional-variance constraints. Corrected log likelihoods above `-150` are flagged for review but are not excluded by this threshold. This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage for this selected model.

> **Note**
>
> Flagged 106 estimate(s) with corrected log likelihood above `-150` for manual review. These rows remain eligible if they pass the admissibility checks.

20.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC	Max Shape	Max Implied Var	Above −150 Diagnostic
ARX(2,2)	4	20	−37.4525	100.9050	144.7233	380.8158	97.5593	yes

Selection checks:

- Optimizer convergence: `yes`
- Parameter bounds: `yes`
- BEGE stability and variance restrictions: `yes`
- Shape upper-cap diagnostic: `flagged`
- Implied variance bounds: `yes`
- Mean-process stationarity: `yes`
- Selection diagnostics: `eligible`
- Standard errors: `computed`

Mean process:

$$\pi_{t+1} = \underset{(0.0000)}{0.0018} + \underset{(0.0000)}{0.2393}\,\pi_t + \underset{(0.0000)}{0.3107}\,\pi_{t-1} + \underset{(0.0054)}{-0.4637}\,SPF_t + \underset{(0.0054)}{0.0534}\,SPF_{t-1} + u_{t+1} \tag{74}$$

BEGE volatility process:

$$\begin{gathered} u_t = \underset{(0.0000)}{0.1808}\,\omega_{p,t} - \underset{(0.0000)}{1.8998}\,\omega_{n,t}, \\ \omega_{p,t} \sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{gathered} \tag{75}$$

$$
\begin{aligned}
p_t &= \underset{(0.0000)}{6.4515} + \underset{(0.0000)}{0.9551}\, p_{t-1} + \frac{\underset{(0.0000)}{0.0559}}{2\left(\underset{(0.0000)}{0.1808}\right)^2} \left(u_{t-1}^{+}\right)^2, \\
n_t &= \underset{(0.0000)}{0.1348} + \underset{(0.0000)}{0.9943}\, n_{t-1} + \frac{\underset{(0.0000)}{0.0000}}{2\left(\underset{(0.0000)}{1.8998}\right)^2} \left(u_{t-1}^{-}\right)^2
\end{aligned}
\tag{76}
$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0018	0.0000
rho_1	0.2393	0.0000
rho_2	0.3107	0.0000
phi_1	−0.4637	0.0054
phi_2	0.0534	0.0054
p0	6.4515	0.0000
n0	0.1348	0.0000
rho_p	0.9551	0.0000
rho_n	0.9943	0.0000
phi_p_plus	0.0559	0.0000
phi_n_minus	0.0000	0.0000
sigma_p	0.1808	0.0000
sigma_n	1.8998	0.0000

21. Full BEGE

BEGE GARCH has mean process and higher:

$$\begin{aligned} \pi_{t+1} &= \hat{\pi}_{t+1} + u_{t+1}, \\ u_{t+1} &= \sigma_p \, \omega_p - \sigma_n \, \omega_n, \end{aligned} \tag{77}$$

where

$$\omega_p \sim \tilde{\Gamma}(p_t, 1), \quad \omega_n \sim \tilde{\Gamma}(n_t, 1). \tag{78}$$

Here n_t, p_t is generated recursively through a GJR-type updating equation with parameters $(p_0, n_0, \rho_p, \rho_n, \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^-)$:

$$\begin{aligned} p_t &= p_0 + \rho_p \, p_{t-1} + \frac{\phi_p^+}{2\sigma_p^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_p^-}{2\sigma_p^2} \left(u_{t-1}^-\right)^2, \\ n_t &= n_0 + \rho_n \, n_{t-1} + \frac{\phi_n^+}{2\sigma_n^2} \left(u_{t-1}^+\right)^2 + \frac{\phi_n^-}{2\sigma_n^2} \left(u_{t-1}^-\right)^2 \end{aligned} \tag{79}$$

The parameter bounds are chosen to be:

- $0 \leq p_0, n_0 < 10$,
- $0 \leq \rho_p, \rho_n \leq 1$,
- $0 \leq \phi_p^+, \phi_p^-, \phi_n^+, \phi_n^- \leq 2$,
- $10^{-5} < \sigma_p, \sigma_n < 2$.

According to the emails, I have set the constraints below.

- $\rho + \frac{\phi^+}{2} + \frac{\phi^-}{2} < 1$ for both BE and GE shape processes.
- $\sigma_p^2 p_0 + \sigma_n^2 n_0 < \mathrm{Var}(\pi_t)$ where $\mathrm{Var}(\pi_t) = 0.75$ is the unconditional variance of inflation.
- The implied variance path $\sigma_p^2 p_t + \sigma_n^2 n_t$ must satisfy the project EWMA lower and upper bounds at every effective-sample observation.

The hard cap $\max\{p_t, n_t\} < 200$ is not imposed during raw multi-start optimization or reported best-model selection by default. The stabilized BEGE density is evaluated directly as long as the recursive shape series are finite; large shape states use the saddlepoint density backend. The collector writes the maximum shape path as a diagnostic in `results/selection_diagnostics.csv`.

22. Batch Estimation

`BEGE_GJR1.py` estimates the Full BEGE specification on the canonical effective sample from `DataSummary/README.md`, **1969Q2–2022Q4** with **215 observations**. The runner uses the precomputed lag columns when they are available, so lagged ARX regressors do not trim the sample again.

The script estimates the four mean-process specifications:

- Constant
- ARX(1,1)
- ARX(2,1)
- ARX(2,2)

Results are checkpointed to `output/raw/draw_###.csv` after each draw. This keeps completed mean-process rows when a seed job is interrupted, while the final write still contains all requested rows when the seed finishes.

`collect_full_results.py` merges the raw seed files and writes:

- `results/all_estimations.csv`, without standard-error columns or optimizer messages.
- `results/by_mean/constant.csv`, `results/by_mean/ARX_1_1.csv`, `results/by_mean/ARX_2_1.csv`, and `results/by_mean/ARX_2_2.csv`, which split the cleaned estimations by mean process.
- `results/selection_diagnostics.csv`, with stored and corrected likelihood criteria plus selection diagnostics.
- `results/best_loglik_with_se.csv`, with standard errors for the likelihood-best admissible fit.
- `results/best_model.md`, with only the likelihood-best admissible fit, reported as substituted mean and BEGE volatility equations with standard errors shown below the parameter values.

When `START_ID` and `END_ID` are set, the collector only merges `draw_###.csv` files in that seed range. This prevents older seed files from entering a new smaller resubmission.

`SimulationGJR.sh` defaults to 50 seed jobs, 40 draws per mean process, 25 optimizer starts per draw, and 800 SLSQP iterations per start. This gives 50,000 optimizer starts for each mean process and Full BEGE specification. With all four mean processes, that is 200,000 optimizer starts total.

23. Full BEGE Best Model Summary

Generated: `2026-06-03T14:41:38` Total estimations: `8000` Converged estimations: `8000` Eligible estimations for best-model selection: `7999`

Saved likelihoods are recomputed from the stored parameter paths before ranking. Large recursive shape states are evaluated by the BEGE saddlepoint density backend; `max(p_t, n_t)` is reported as a diagnostic, not as an exclusion rule.

Selection screen: finite corrected AIC/BIC/log-likelihood, successful optimizer status, finite positive shape paths, positive conditional variance paths, EWMA implied-variance bounds, mean-process stationarity, and documented parameter/stability/unconditional-variance constraints. Corrected log likelihoods above `-150` are flagged for review but are not excluded by this threshold. This report shows only the single likelihood-best admissible estimate. Standard errors are computed at the reporting stage for this selected model.

Note

Flagged 104 estimate(s) with corrected log likelihood above `-150` for manual review. These rows remain eligible if they pass the admissibility checks.

23.a. Selected Best Model

Best admissible estimate ranked by corrected log likelihood.

Mean	Seed	Draw	LogLik	AIC	BIC	Max Shape	Max Implied Var	Above −150 Diagnostic
ARX(2,2)	16	1	41.1967	−52.3934	−1.8339	410.2149	30.8171	yes

Selection checks:

- Optimizer convergence: `yes`
- Parameter bounds: `yes`
- BEGE stability and variance restrictions: `yes`
- Shape upper-cap diagnostic: `flagged`
- Implied variance bounds: `yes`
- Mean-process stationarity: `yes`
- Selection diagnostics: `eligible`
- Standard errors: `computed`

Mean process:

$$\pi_{t+1} = \underset{(5.8189)}{0.0538} + \underset{(88.3952)}{0.2275}\,\pi_t + \underset{(16.7431)}{0.0073}\,\pi_{t-1} + \underset{(2328.5311)}{0.7791}\,SPF_t + \underset{(1925.4941)}{0.2126}\,SPF_{t-1} + u_{t+1} \quad (80)$$

BEGE volatility process:

$$\begin{aligned} u_t &= \underset{(0.0190)}{0.0466}\,\omega_{p,t} - \underset{(28.6017)}{0.2734}\,\omega_{n,t}, \\ \omega_{p,t} &\sim \tilde{\Gamma}(p_t, 1), \qquad \omega_{n,t} \sim \tilde{\Gamma}(n_t, 1). \end{aligned} \quad (81)$$

$$
\begin{aligned}
p_t &= \underset{(996.7706)}{5.1586} + \underset{(1.1682)}{0.9061}\, p_{t-1} + \frac{\underset{(1.5762)}{0.0101}}{2\left(\underset{(0.0190)}{0.0466}\right)^2}\left(u_{t-1}^{+}\right)^2 + \frac{\underset{(0.0000)}{0.0373}}{2\left(\underset{(0.0190)}{0.0466}\right)^2}\left(u_{t-1}^{-}\right)^2, \\
n_t &= \underset{(40.8905)}{6.2259} + \underset{(0.3850)}{0.8302}\, n_{t-1} + \frac{\underset{(58.1956)}{0.1150}}{2\left(\underset{(28.6017)}{0.2734}\right)^2}\left(u_{t-1}^{+}\right)^2 + \frac{\underset{(31.2553)}{0.1942}}{2\left(\underset{(28.6017)}{0.2734}\right)^2}\left(u_{t-1}^{-}\right)^2
\end{aligned}
\tag{82}
$$

Parameter table:

Parameter	Estimate	Std. Error
c	0.0538	5.8189
rho_1	0.2275	88.3952
rho_2	0.0073	16.7431
phi_1	0.7791	2328.5311
phi_2	0.2126	1925.4941
p0	5.1586	996.7706
n0	6.2259	40.8905
rho_p	0.9061	1.1682
rho_n	0.8302	0.3850
phi_p_plus	0.0101	1.5762
phi_p_minus	0.0373	0.0000
phi_n_plus	0.1150	58.1956
phi_n_minus	0.1942	31.2553
sigma_p	0.0466	0.0190
sigma_n	0.2734	28.6017